INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

WELDING MACHINES, ELECTRICAL RESISTANCE, ROLL SPOT AND SEAM, PRESS TYPE, THREE PHASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three phase, press type, roll spot and seam, electrical resistance welding machines of the direct energy, low-frequency converter or rectifier direct current, air actuated type, having the capability to produce roll spot and seam welds with precision controls on ferrous and non-ferrous material.

1.2 Classification. The specification covers machines of the following types, sizes and classes. The type, size and class to be furnished shall be as specified. (See 6.2.1).

Type I - Low Frequency Converter
Type II - Rectifier Direct Current

Size 1 - 50 kva transformer rating
Size 2 - 100 kva transformer rating
Size 3 - 200 kva transformer rating
Size 4 - 300 kva transformer rating
Size 5 - 400 kva transformer rating

Class A - 36 inch throat depth
Class B - 48 inch throat depth
Class C - 54 inch throat depth
Class D - 60 inch throat depth

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Throat dimensions. The welding machine throat dimension is the unobstructed distance, measured in the welding plane, from the centerline welding electrodes to the first obstruction of the machine frame.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

MILITARY

MIL-M-6858 - Welding, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-resisting Alloys, and Titanium Alloys, Spot and Seam.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be that in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

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29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 93.114M - Pneumatic Fluid Power Systems. Standard for Industrial Equipment and Machine Tool Applications

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards. (DOD adopted)

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provide such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for maximum energy efficiency as governed by the developments available at the time of manufacture within the industry.

3.2.4 Controls. All mechanical, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of 29, CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2.1). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

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3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solder or paste shall not be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4. Components. The machine shall consist of not less than a frame and base, enclosure, upper electrode drive head, lower electrode head, electrode/wheels, upper ram and head assembly, lower knee assembly, electrical system, pneumatic system, and cooling system.

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3.4.1 Frame and base. The frame and base assembly shall be a steel weldment of box construction, ribbed to maintain original alignment and to minimize deflection. The base shall provide for floor mounting. Access to the interior shall be provided to permit inspection and servicing of the equipment installed within the frame. The front frame shall be designed to permit positioning of the upper arm and head assembly and the lower knee and horn assembly.

3.4.2 Enclosure. The metal enclosure shall be constructed of either corrosion resistant material or metal treated to resist corrosion. Ventilation features shall be adequate to prevent temperatures in excess of the design requirements under normal operating conditions of rated power output.

3.4.3 Upper electrode drive head. The upper holder or electrode head, normally the drive head, shall be of the universal type, electrically driven and rotatable 90 degrees to the right or left to permit circumferential or longitudinal welding when the machine is equipped with the optional lower electrode and arm assembly (see 3.4.4). The wheel shaft shall be supported at both ends by anti-friction bearings, maintaining positive wheel alignment. The welding current shall be conducted from the stationary housing to the rotating wheel shaft by a permanent type bearing as an independent function from the bearing carrying the thrust load and transmitting the electrode force to the welding wheel. The wheel shall be driven without interfering with the working area. Maximum backlash of the drive system shall not exceed 0.020 inch measured at the periphery of a twelve inch wheel. Means shall be included for both continuous and intermittent drive. The electrode wheel shaft speed shall be infinitely variable or adjusted in increments of 1 percent steps from 0 to 99 percent of maximum wheel speed. The values of time of drive, and time of stop shall be incrementally adjustable without removing or replacing components.

3.4.4 Lower electrode head. The lower electrode head shall be the idling type. The electrode head and arm shall be integrated to provide maximum work area levels for the welding wheel. The force and thrust loads shall be independent from the current conduction function as specified in 3.4.3. When specified (see 6.2.1) conventional sleeve bearings will be acceptable when the clearance limitations of the welding application precludes the use of the former type electrode head for longitudinal welding. When specified (see 6.2.1) the lower electrode head and arm assembly shall be of the longitudinal or both longitudinal and circumferential welding type.

3.4.5 Electrode/wheels. Unless otherwise specified (see 6.2.1) two internally cooled electrode/wheels of equal diameter shall be furnished with the machine. The wheels shall be made of a copper base alloy conforming to the standards of the RWMA Bulletin Number 16. The diameter, alloy class and quantity of additional electrode/wheels shall be as specified (see 6.2.1). Special or unequal diameter wheels shall be as specified (see 6.2.1).

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3.4.6 Upper ram and head assembly. The upper ram and head assembly shall be constructed to maintain its original alignment and to meet or exceed the deflection requirements specified in table I. The guiding mechanism for the ram shall be of the anti-friction type, provided with adjustments for wear, and protected against dust and foreign matter. The machine shall be equipped with a retractable head, controlled by a foot switch. The work stroke, electrode force, and the total retraction stroke shall be as shown in table II. The cylinder/diaphragm shall be cushioned for shockless operation.

3.4.7 Lower knee assembly. The lower knee shall provide a vertical adjustment range of not less than eleven inches. A hydraulic jack shall be permanently mounted on the base of the machine to aid in the adjustment of the knee. With maximum rated electrode force applied and the lower knee assembly unsupported by any bracing means, the difference between the horizontal deflection of the upper electrode/wheel and the horizontal deflection of the lower electrode/wheel shall not exceed the value specified in table IT The vertical deflection of the lower electrode/wheel shall not exceed the value specified in table III.

3.4.8 Electrical system. The electrical system shall include the welding transformer(s), rectifier(s), timer(s), suitable circuit protection, switches and all controls as required for installation and operation. The electrical system and components shall conform to the requirements of NFPA 79. All interconnects between panels shall be made with screw-type terminal strips or Plugs. All leads to screw-type terminals shall end in solderless terminals. All interconnection and control panel wiring shall be accomplished with conductors of 20 gage AWG or larger stranded copper wire, and shall be color or tag coded. All plug-in devices shall be fastened with suitable clamps or retainers. Each sub-panel shall be arranged so that they may be removed independently of any other sub-panel. All terminals and units shall be clearly identifiable from the machine electrical wiring diagram. Replaceable filters shall be provided when forced air is applied for ventilation.

TABLE I. Horizontal deflection difference between the upper electrode and the lower electrode, maximum allowable.

Machine Size	Difference (inches)
1	0.0035
2	0.0040
3	0.0045
4	0.0050
5	0.0055

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TABLE II. Machine size and principal characteristics.

Machine Size	Welding transformer KVA rating at 50% Duty cycle	Secondary Current at short circuit condition range Type I Machine (AMPS)		Secondary Welding current range Type II Machine (AMPS)		Electrode force range at 80 PSI air pressure (LBS)		Stroke (inches)		Electrode wheels thickness inches
								Working retraction		
		MIN.	MAX.	MIN.	MAX.	MIN.	MAX.			
1	50	4,500	45,000	4,500	45,000	120	1,200	1/2	2 1/2	1/2
2	100	6,000	60,000	5,500	55,000	260	2,600	1/2	2 1/2	3/4
3	200	10,000	100,000	8,000	80,000	400	4,000	5/8	3 3/8	3/4
4	300	13,000	130,000	11,000	110,000	600	6,000	3/4	4	1
5	400	15,000	150,000	12,500	125,000	1,100	11,000	1	6	1 1/4

NOTE: Where a range is shown, requirements shall range from not more than the minimum to not less than the maximum figures shown in the table.

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TABLE III. Vertical deflection of the lower electrode holder, maximum allowable

Machine Size	Throat Depth (inches)			
	36	48	54	60
1	.031	.033	.034	.035
2	.029	.031	.032	.033
3	.027	.029	.030	.031
4	.025	.027	.028	.029
5	.023	.025	.026	.027

3.4.8.1 Input electrical power. The welding machines shall be operable on a 60 hertz, essentially balanced, three-phase primary electric service at one of the following voltages; 208, 230, 460, or 575 volts. Unless otherwise specified (see 6.2.1), the welding machine shall be initially wired for operation on 460 volts.

3.4.8.2 Main disconnect circuit breaker. The machine shall have a main disconnect circuit breaker. Unless otherwise specified (see 6.2.1) a circuit breaker shall be furnished with an under voltage trip coil and shall be mounted on the machine panel. The circuit breaker shall provide a safety interlock when access door(s) are opened to areas that are a hazard to the operator.

3.4.8.3 Welding Transformer. The welding transformer shall be mounted on the machine. The transformer shall be of the low frequency conversion type for the type I or the rectifier type for the type II. The transformer shall operate on an essentially balanced three-phase power supply at a vector power factor of not less than 85 percent. The transformer KVA rating and secondary current shall be in accordance with table II for the machine specified. The transformer shall be water cooled and thermostatically controlled to minimize condensation. A temperature responsive device shall be provided to prevent energization of the transformer if the temperature of the transformer exceeds design limits. All connecting surfaces of secondary circuit shall be silver plated. The transformer shall be insulated to conform to RWMA Bulletin Number 16 for class B or better insulation. When specified (see 6.2.1), the transformer shall have a series parallel switch in the primary circuit. Transformer taps connected to a selector switch shall be provided for the type II machine as necessary to produce the required heat range for the welding.

3.4.8.4 Secondary current rectifier system. The type II machine shall have a secondary current rectifier system. The rectifier system shall be the silicon diode type using three-phase, full wave circuit arrangement or a six

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phase, half wave circuit arrangement. The forward current rating and reverse peak voltage rating of the diodes shall be compatible with the size machine specified. The silicon diodes shall be mounted on a water cooled conductor. Within each phase the impedance of each diode circuit shall be similar, allowing each diode with similar electrical characteristics to load equally. The diodes shall be arranged for installation and removal without damage to the rectifier system. Surge suppression protection shall be provided with sufficient capacity to reduce any transient voltage peak which would cause failure of the diodes from either AC or DC reflected transient voltage. The rectifier system shall have protective control circuitry which will open the line circuit breaker upon excessive temperature in the rectifier system.

3.4.8.5 Electrical controls. The electrical controls shall consist of the welding power control, timing and sequence controls, welding contactor, current adjustment control, and auxiliary and accessory controls. All calibrations, phase shift controls, and similar characteristics shall be clearly marked in standard terms such as: cycles (cy.), impulses (imp.), percent of maximum current (% max.).

3.4.8.5.1 Operator control panel. The operator control panel shall have an on-off switch or weld/no switch, sequence lights, a current adjustment control, a weld timer selector, auxiliary and accessory control function switches, and any other controls or switches required to meet the requirements specified herein.

3.4.8.5.2 Welding power control unit. The machine shall have a solid state synchronous, precision, three phase, welding power control unit. The control unit may be located either in a separate cabinet, or mounted as an integral part of the machine.

3.4.8.5.3 Welding contactor. The welding contactor shall be a silicon controlled rectifier (SCR) type. The SCR contactor shall be water cooled and thermostatically protected. Means shall be provided to prevent welding if the water is shut off or if the supply is restricted. The SCR contactor shall be provided with over-current and over-voltage protective circuits. Safety relays shall be provided to isolate the SCR's from the control circuit when the machine is in the rest position. The welding contactor and rectifier may be single units operating in the primary circuit of the welding transformer of the low-frequency converter type machines, or they may be separate units operating in the primary and secondary circuits, respectively, of the rectifier direct-current type. On low-frequency converter systems, the welding current output of the positive and negative rectifier shall be inherently balanced and individual controlled inputs shall be provided to select and weld with uni-directional or bi-directional current of one, two or three phases. Lights or read-outs shall indicate firing of individual SCR banks.

3.4.8.5.4 Timing and sequencing controls. The control equipment shall provide operating control and adjustment of the welding variable to ensure that selected values of force and current timing and sequencing can be repetitively attained to produce spot or seam welds conforming to MIL-W-6858. The sequence shall be the non-beat type before or after the squeeze function. The control shall be of the synchronous precision type with solid state

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components. Use of relays shall be kept to an absolute minimum and only such relays as are required to assure safety and reliable operation will be used. Relays within control circuits shall be the plug-in type. The timing and sequencing functions shall be grouped on a plug-in chassis. The control chassis and components shall be mounted to prevent damage to the unit through the service life of the machine. The controls shall have at least the following standard or equivalent timing functions; squeeze, weld, cool, post heat, forge delay, hold, index delay, index and off. When specified (see 6.2.1), the machine shall be equipped with the following additional functions: preheat, quench and temper for welding heat resistant materials. Additional control functions shall be furnished as specified (see 6.2.1). The timing controls shall be calibrated in cycles (cy.) based on the electrical servicing line frequency of 60 hz, except that the preheat, weld, and temper interval control input may be calibrated in either cycles (cy.) or impulses (imp.). The timing controls shall have a minimum range of 0-99 cycles or impulses. When a forge-delay timer is used the timers range 0.1 cycles and accuracy shall be ±.05 cycle. The controls shall have signal lights on the panels to indicate the operating sequence of the welding cycles and to aid in trouble shooting. The timing and sequencing controls shall be selected and adjusted by means of knobs, switches or digital input. The knobs, switches, digital input and lights shall be clearly visible by, and easily accessible to operator.

3.4.8.5.5 Current adjustment control. Unless otherwise specified (see 6.2.1) adjustment of the welding current, from the upper to the lower limits of the machine capacity, shall be provided. The adjustment shall be by means of phase shift heat control, transformer tap switches or digital input selection, or a combination thereof. The control heat shall be graduated either in increments of 1 from 0 to no less than 99 of 20 percent through 99 percent in 1 percent increments. The current adjustment shall be, such that, at the maximum setting of the adjustment input, the maximum short-circuit current for type I machines or secondary welding current for type II machines is produced. At all settings of the controls, the maximum current produced shall be reproducible within 1 percent for constant input conditions. The preheat, weld and temper current levels shall be adjustable independently by means of individual phase controls.

3.4.8.5.6 Automatic voltage compensator. The machine shall have a solid state automatic voltage compensation (AVC) system to minimize the effects of line voltage variations. The AVC shall maintain the welding current within plus or minus 2 percent, for a primary line voltage variation of plus or minus 10 percent. The AVC shall employ 3 phase line voltage sensing. Unless otherwise specified (see 6.2.1), based on the input line frequency, compensation shall be initiated within 1/3 cycle for a sudden change in line voltage in any of the three phases. When applicable (see 3.4.8.5.4) the AVC shall operate on the preheat, weld and temper currents.

3.4.8.5.7 Control circuit voltage. Control equipment shall be capable of operating on control circuit voltage which varies as much as ±10 percent of the voltage for which the equipment is adjusted. All switch inputs shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 24 volts.

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3.4.8.5.8 Worklight. When specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.8.5.9 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the welding machine. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.8.6 Accessory controls. Accessory controls shall consist of at least the following.

3.4.8.6.1 Foot switch. The foot switch shall control the upper electrode assembly and its associated pneumatic system and initiate the welding cycle. A two-stage foot switch shall be supplied which shall initiate the electrode force in the first stage, and initiate the welding sequence in the second stage. The foot switch shall be guarded to prevent accidental operation of the machine.

3.4.8.6.2 Wheel dress control. The wheel dress control shall operate to lower the upper electrode/wheel under a reduced force for cleaning and aligning the electrodes.

3.4.8.6.3 Contact pressure gauge switch. The contact pressure gauge shall be a presettable pressure switch that shall prevent continuation of the welding sequence, as initiated by the foot switch, until the air pressure in the pneumatic head is equal to the contact gauge setting.

3.4.8.6.4 Electrode force interlock circuit. The electrode force interlock circuit shall prevent the upper electrode from being raised during the welding cycle. The weld shall terminate immediately when the retraction switch is released and upon loss of air pressure.

3.4.8.6.5 Weld/no weld (on/off) switch. A weld/no weld switch shall be provided. In the no weld position, the control sequence will actuate except that no welding current shall flow.

3.4.8.6.6 Electrode force selector. The electrode force selector shall permit selection of a constant high, a constant low, and a variable welding force sequence.

3.4.8.6.7 One, two or three phase weld switches. On low-frequency converter systems, individual control switches shall be provided to select and weld with uni-directional or bi-directional welding currents of one, two or three phases. This does not apply to direct current rectifier machines.

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3.4.8.6.8 Electrode wheel drive control. A control convenient to the operator shall be provided to obtain stepless speed adjustment of the electrode wheel rotation. Electronic control of the spacing between roll spot welds shall be obtained by a synchronous control in 1 cycle steps. The control shall maintain accurate weld spacing. The motor control shall provide means to synchronize the initiation of the welding current, immediately after the wheel stops, and to excite the drive in synchronization with the weld cycle.

3.4.8.6.9 Circuit testing. Means shall be provided for circuit testing the control to aid in maintenance checking. The testing shall consist of operating the control through its sequence without welding and without having to energize the foot pedal and operating the control through its sequence one function at a time and cycle by cycle.

3.4.9 Pneumatic system. The pneumatic system and components shall conform to ANSI B93.114M and the following requirements:

3.4.9.1 Pneumatic system operation. The pneumatic system air cylinder/diaphragm shall be a double action type, and shall operate on 80 PSI input air line pressure. The inlet connection shall be marked for the pressure requirements. The pneumatic system shall include valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring, switches, indicator lights, and all related equipment necessary to permit use of any one of three electrode forces (Constant Low, Constant High, or Variable). Air cushions or air speed valves shall insure shockless operation. The system shall be controlled by a solenoid valve actuated by a foot switch. A means for lowering the upper electrode for cleaning and aligning shall be provided. When specified (see 6.2.1), adjustable low force electrode cleaning and aligning shall be provided.

3.4.9.1.1 Pneumatic variable force mode. The variable force mode shall consist of the sequential application of a pneumatic squeeze or precompression force, a weld force, and a forge force. During the variable force sequence, the forging force shall be applied after a timer delay which shall be adjustable and predetermined with respect to the start of the rise of the welding current. The rise of the electrode force from its weld force level to its forge force level, shall be 75 percent accomplished in no more than:

10 milliseconds, for machine sizes 1 and 2
20 milliseconds, for machine sizes 3 and 4
30 milliseconds, for machine size 5

3.4.9.1.2 Air gauges. Air gauges shall have a face diameter of not less than 4-1/2 inches, and shall be calibrated in pounds per square inch (PSI) or pounds of applied electrode force where specified. Gauge accuracy shall be within ±0.5 percent of the maximum reading of the gauge. Means of shock protection shall be provided to prevent gauge damage. The gauge shall be visible to the operator.

3.4.9.1.3 Air regulators. The regulators shall be self-bleeding. Regulators and solenoid valves shall be installed with unions. The air-piloted mainstream regulators shall duplicate the input signal from the pilot

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regulator to within +0.5 PSI throughout the full range of 2 to 100 PSI. The regulators shall hold the output pressure within +0.125 PSI as long as the supply lines remain at least 2 PSI higher than the preset pressure on the regulator. The regulators shall automatically sense and relieve output pressure differences of more than 0.125 PSI from the preset value throughout the full range of 2 to 100 PSI.

3.4.10 Water cooling system. The machine shall have a cooling system that will prevent any temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply, and shall have a single inlet connection. The inlet connection shall be marked for pressure and flow requirements. The system shall have parallel flows from the manifold to provide proper cooling of the welding transformer, rectifiers, contactors, conductors and electrodes. The piping in the system shall be corrosion-resistant in tap water service. The system shall have an inlet strainer, and a closed manifold drain system with flow indicators. The water flow lines to the electrodes shall have solenoid valves (installed with pipe union connections) to prevent water flow while changing the electrodes. When specified (see 6.2.1), the system shall have water saving devices.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table II for the machine size specified.

3.6 Performance. The welding machines shall produce welds conforming to the requirements of MIL-W-6858 and tables I, II, III and IV.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. All standard equipment normally furnished with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in bold face on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. When specified (see 6.2.1), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2.1), the information provided on the plate shall be as listed below.

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TABLE IV. Machine welding capacity two pieces of equal thickness (inches).

Machine Size	Aluminum and magnesium Alloys		Stainless Steels (300 Series) and Titanium Alloys		Low Carbon Steel		Ninomic 75 Ninomic 80 Inconel 600		Rene 41 Inconel 718 Hayes 25	
	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.
1	.010	.025	.008	.033	.008	.048	.008	.027	.005	.012
2	.018	.040	.012	.060.	.012	.075	.012	.036	.007	.018
3	.020	.064	.018	.091	.018	.120	.018	.060	.008	.033
4	.040	.091	.036	.125	.036	.164	.033	.090	.012	.075
5	.051	.125	.054	.164	.060	.194	.048	.120	.018	.105

NOTE: Where a range is shown, requirements shall range from not more than the minimum to not less than the maximum figures shown in the table.

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Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in 4.1.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.7, The tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. When required (See 6.2.1), an acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^0 to 104^0 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational welding test. The machine shall be connected to the specified source of electric, air and water supply and prepared for welding in accordance with the manufacturer's instruction. The machine shall be operated at no load for a minimum of 15 minutes, during which time all functions shall be checked for specified operation. There shall be no malfunction. The machine shall then be tested for welding capability in accordance with MIL-W-6858, with the following exceptions: not less than 30 welds shall be made instead of 105; 25 welds instead of 100 shall be tested in shear with the joints unrestricted; and 5 welds shall be sectioned for metallurgical examination. The test material shall be 2024T3 or 7075T6 alclad aluminum alloy of the maximum thickness specified in table II. If in any one operation the electrodes arc upon separation, the equipment shall be considered as having failed the test in which the arcing occurred.

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4.8.3 Performance welding test for class A welds. The machine shall be tested for welding capability by performing all the tests in MIL-W-6858 for class A welds. The test material shall be 2024T3 or 7075T6 alclad aluminum alloy and stainless steel, type 302 or 304. When this test is specified, the aluminum welding test in 4.8.2 shall not be required. Tests shall be made using each of these materials in the maximum and minimum thickness specified in table IV according to machine size. In any of the above welding tests; if the electrodes arc upon separation, the equipment shall be considered as having failed the test in which arcing occurred.

4.8.4 Electrode force test. The machine shall be tested for electrode force using an electrode force gage. The measured force shall comply with table II.

4.8.5 Electrode deflection test. The machine shall be tested to determine the vertical and horizontal deflection of the upper and lower electrodes. The electrodes shall be positioned at the nominal throat depth location, and the pneumatic system adjusted to provide the maximum electrode force value specified in table I. Without the lower arm or horn being supported by a brace or any other device that can in effect serve as a brace, the electrodes shall be brought to bear upon a plate and roller assembly similar to figure 1A. Under these conditions the difference between the upper electrode and lower electrode horizontal deflection shall not exceed the value specified in table I, and the vertical deflection of the lower electrode holder shall not be more than that specified in table III.

4.8.6 Secondary current test. As applicable, the maximum and minimum wheel to wheel short-circuit secondary current of the type I machine or the secondary welding current of the type II machine shall be determined. The minimum and maximum values thus determined shall comply with the values specified in table II.

4.8.7 Forge force rise time. With the machine sequence set in the variable force mode, and with the machine controls adjusted to the proper values for welding two pieces of the maximum thickness of 2024T3 or 7075T6 alclad aluminum specified in table IV in accordance with MIL-W-6858, the rise time of the forge force and the time relationship of the weld current and the forge force shall be measured by means of a calibrated recording instrument. The rise of the electrode force in the variable pressure mode from its weld force level to its forge force level shall be 75 percent accomplished in no more than the time specified in 3.4.9.1.1. The calibration accuracy of the forge delay timer shall comply with 3.4.8.5.4.

4.8.8 Automatic voltage compensator test. The performance of the automatic voltage compensator shall be measured by means of a calibrated recording instrument. With the current adjustment set first at 50 percent and then at 90 percent of maximum, the secondary current and primary voltage shall be recorded as ±10 percent step changes are introduced into the primary voltage while secondary current is flowing. Alternately, the compensation and response time may be tested by causing the voltage change to occur only on the control of the compensator, in which case, the 10 percent increase or decrease in voltage shall cause, respectively, a decrease or increase of 10 percent in the secondary current within ± 2 percent. The secondary current compensation and response time shall be as specified in 3.4.8.5.6.

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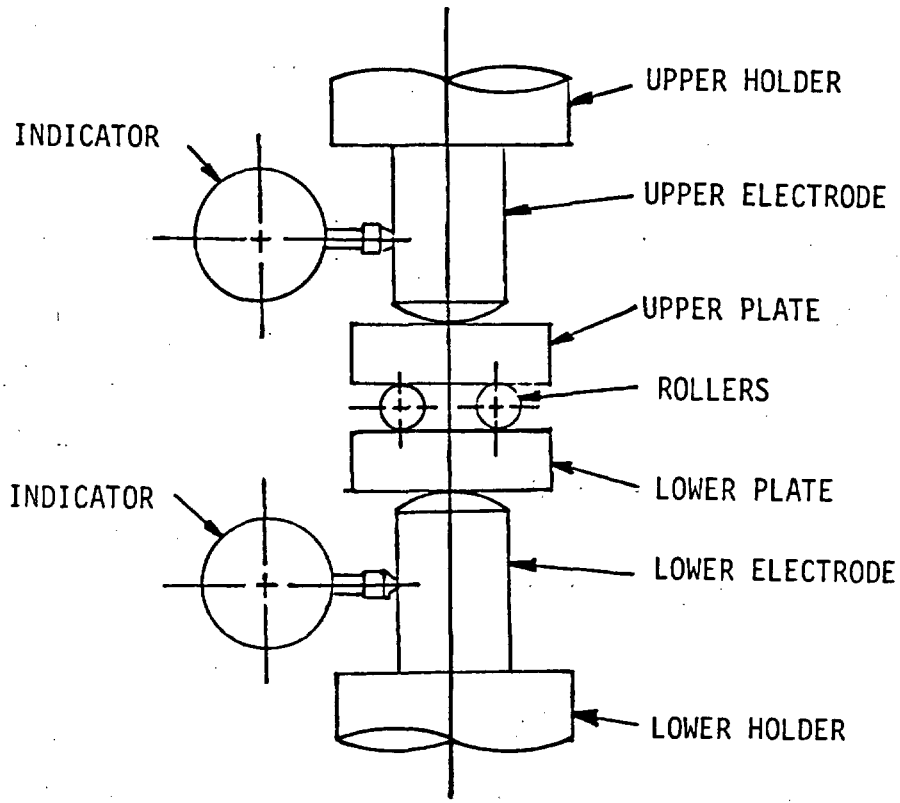


FIGURE 1A. DETAIL OF DIAL INDICATOR SET-UP FOR MEASURING ELECTRODES HORIZONTAL DEFLECTION.

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4.8.9 Dielectric test. The machine shall be subjected to the dielectric test described in RWMA Bulletin 16.

4.8.10 Rated capacity tests. The machine shall be adjusted to the manufacturers recommended settings for the maximum thickness of aluminum specified for the machine being tested. The machine shall be tested under continuous operation for 1 hour, simulating seam welding, except that the two electrodes shall come into direct contact. External cooling of the electrodes by flooding with water, in addition to the internal cooling, will be permitted to compensate for the heat normally dissipated in the workpiece passing between the electrodes. The temperature protective devices shall not trip and there shall be no evidence of damage to the machine. If the equipment fails to conform to any of these requirements, or fails to operate properly during the test, the test shall be discontinued, and the supplier permitted to make any necessary adjustments. After adjustment, the test in which the equipment failed shall be completely rerun. Failure to pass the rerun test shall be cause for rejection.

4.8.11 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.4 shall be as specified and fully described (see 6.2.1).

4.8.12 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The welding machines covered by this specification are intended for use in producing class A resistance welds under production conditions in accordance with this specification and the requirements of MIL-W-6858, as applicable, in the materials and thickness indicated herein. The welding machines are for broad and diverse usage for aircraft, missile, and spacecraft, including the welding of critically loaded primary structural assemblies.

6.2 Acquisition requirements.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5.).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

j. Sleeve bearing, if acceptable (see 3.4.4).

k. Lower electrode head and arm assembly, for longitudinal or both longitudinal and circumferential welding, if required (see 3.4.4).

l. Diameter, alloy class and quantity of electrodes/wheels required (see 3.4.5).

m. Special electrode wheel, if required (see 3.4.5).

n. Voltage connection required (see 3.4.8.1).

o. Circuit breaker and its mounting, if different (see 3.4.8.2).

p. Series parallel switch, if required (see 3.4.8.3).

q. Preheat, weld and temper control functions, if required (see 3.4.8.5.4).

r. Additional control functions, if required (see 3.4.8.5.4).

s. Current adjustment control, if different (see 3.4.8.5.5).

t. Automatic voltage compensator, if different (see 3.4.8.5.6).

u. Worklight, if required (see 3.4.8.5.8).

v. Hour meter, if required (see 3.4.8.5.9).

w. Adjustable low force, if required (see 3.4.9.1).

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x. Water saving devices, if required (see 3.4.10).

y. Size and capacity, if different (see 3.5).

z. Additional performance features, if required (see 3.6.1).

aa. Specify and fully describe optional equipment, if required (see 3.8).

bb. Specify applicable repair parts, if required (see 3.9).

cc. Lubrication plate, if required (see 3.10.1).

dd. Nameplate, if different (see 3.10.2).

ee. Inspection conditions, if different (see 4.3).

ff. First article inspection, if different (see 4.4).

gg. Quality conformance inspection, if different (see 4.5).

hh. Acceptance test, if different (see 4.6).

ii. Additional performance tests (see 4.8.11).

6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.2.1 Technical data. All technical data furnished should be written in the English language.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the

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number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Cross-reference of classification Classification changes from the previous issue of this specification are as follows:

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

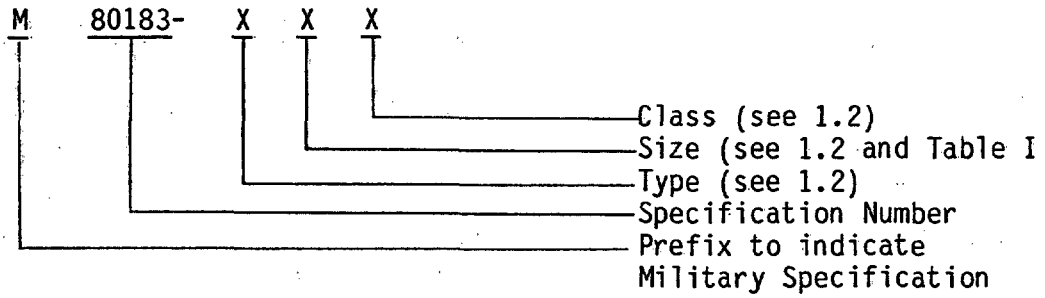
6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Electrode wheels
Lower arm
Pneumatic
Rectifier
Upper arm
Welding transformer

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 80, 84

User activities:
Navy - MC, CG, OS

Preparing activity:
DLA - IP

Project (3432-0140)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80183C	Welding Machines, Electrical Resistance, Roll Spot and Seam, Press Type, Three Phase

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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